

	РБНФ №2 (опис формальна граматика мови)	Опис формальної граматики мови	Опис граматики з специфікацією lookahead для LL2-аналізатора	/* Код для перевірки РБНФ №1 */	/* Код для перевірки РБНФ №2 */	/* Дані для LL2-аналізатора (лексеми вже опрацьовані лексичним аналізатором) УВАГА: при копіюванні зважайте, щоб у кожному рядку після символу «\» не містилось жодних інших символів. */
РБНФ №1 (опис синтаксису мови всіма допустимими засобами РБНФ)		G = (N, T, P, S)	G = (N, T, P, S)			
		S → program_rule	S → program_rule			
		N = { labeled_point, Ident, goto_label, program_name, value_type, array_specify, declaration_element, array_specify__optional, other_declaration_ident, declaration, other_declaration_ident__iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, value, index_action__optional, expression, binary_action__iteration, expression_or_cond_block__with_optional_assign, assign_to_right, assign_to_right__optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else__iteration, body_for_false__optional, cycle_begin_expression, cycle_end_expression, cycle_counter, cycle_counter_lr_init, cycle_counter_init, cycle_counter_last_value, cycle_body, forto_direction, forto_cycle, continue_while, break_while, statement_in_while_and_if_body, statement, block_statements_in_while_and_if_body, statement_in_while_and_if_body__iteration, while_cycle_head_expression, while_cycle, repeat_until_cycle_cond, repeat_until_cycle, statements__or__block_statements, block_statements, input_rule, argument_for_input, output_rule, statement__iteration, expression__optional, program_rule, declaration__optional, unsigned_value, value, sign__optional, sign, ident, sign_plus, sign_minus }	N = { labeled_point, Ident, goto_label, program_name, value_type, array_specify, declaration_element, array_specify__optional, other_declaration_ident, declaration, other_declaration_ident__iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, value, index_action__optional, expression, binary_action__iteration, expression_or_cond_block__with_optional_assign, assign_to_right, assign_to_right__optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else__iteration, body_for_false__optional, cycle_begin_expression, cycle_end_expression, cycle_counter, cycle_counter_lr_init, cycle_counter_init, cycle_counter_last_value, cycle_body, forto_direction, forto_cycle, continue_while, break_while, statement_in_while_and_if_body, statement, block_statements_in_while_and_if_body, statement_in_while_and_if_body__iteration, while_cycle_head_expression, while_cycle, repeat_until_cycle_cond, repeat_until_cycle, statements__or__block_statements, block_statements, input_rule, argument_for_input, output_rule, statement__iteration, expression__optional, program_rule, declaration__optional, unsigned_value, value, sign__optional, sign, ident, sign_plus, sign_minus }	#define NONTERMINALS labeled_point,\ Ident,\ goto_label,\ program_name,\ value_type,\ array_specify,\ declaration_element,\ \ other_declaration_ident,\ declaration,\ \ index_action,\ unary_operator,\ unary_operation,\ binary_operator,\ binary_action,\ left_expression,\ group_expression,\ value,\ \ expression,\ \ expression_or_cond_block__with_optional_assign,\ assign_to_right,\ \ if_expression,\ body_for_true,\ false_cond_block_without_else,\ body_for_false,\ cond_block,\ \ cycle_begin_expression,\ cycle_end_expression,\ cycle_counter,\ cycle_counter_lr_init,\ cycle_counter_init,\ cycle_counter_last_value,\ cycle_body,\ forto_direction,\ forto_cycle,\ continue_while,\ break_while,\ statement_in_while_and_if_body,\ statement,\ block_statements_in_while_and_if_body,\ \ while_cycle_head_expression,\ while_cycle,\ repeat_until_cycle_cond,\ repeat_until_cycle,\ statements__or__block_statements,\ block_statements,\ input_rule,\ argument_for_input,\ output_rule,\ \ \ program_rule,\ \ unsigned_value,\ value,\ \ sign,\ ident,\ sign_plus,\ sign_minus	#define NONTERMINALS labeled_point,\ Ident,\ goto_label,\ program_name,\ value_type,\ array_specify,\ declaration_element,\ array_specify__optional,\ other_declaration_ident,\ declaration,\ other_declaration_ident__iteration,\ index_action,\ unary_operator,\ unary_operation,\ binary_operator,\ binary_action,\ left_expression,\ group_expression,\ value,\ index_action__optional,\ expression,\ binary_action__iteration,\ expression_or_cond_block__with_optional_assign,\ assign_to_right,\ assign_to_right__optional,\ if_expression,\ body_for_true,\ false_cond_block_without_else,\ body_for_false,\ cond_block,\ false_cond_block_without_else__iteration,\ body_for_false__optional,\ cycle_begin_expression,\ cycle_end_expression,\ cycle_counter,\ cycle_counter_lr_init,\ cycle_counter_init,\ cycle_counter_last_value,\ cycle_body,\ forto_direction,\ forto_cycle,\ continue_while,\ break_while,\ statement_in_while_and_if_body,\ statement,\ block_statements_in_while_and_if_body,\ statement_in_while_and_if_body__iteration,\ while_cycle_head_expression,\ while_cycle,\ repeat_until_cycle_cond,\ repeat_until_cycle,\ statements__or__block_statements,\ block_statements,\ input_rule,\ argument_for_input,\ output_rule,\ statement__iteration,\ expression__optional,\ program_rule,\ declaration__optional,\ unsigned_value,\ value,\ sign__optional,\ sign,\ ident,\ sign_plus,\ sign_minus	
		T = { " ,", "GOTO", "INTEGER16", " ,", "NOT", "AND", "OR", "==", "!=", "<=", ">=", "<", ">", "+", "-", "*", "/", "MOD", "(", ")", "=", "ELSE", "IF", "DO", "FOR", "TO", "DOWNTO", "WHILE", "CONTINUE", "BREAK",	T = { " ,", "GOTO", "INTEGER16", " ,", "NOT", "AND", "OR", "==", "!=", "<=", ">=", "<", ">", "+", "-", "*", "/", "MOD", "(", ")", "=", "ELSE", "IF", "DO", "FOR", "TO", "DOWNTO", "WHILE", "CONTINUE", "BREAK",	#define TOKENS tokenCOLON\ tokenGOTO\ tokenINTEGER16\ tokenCOMMA\ tokenNOT\ tokenAND\ tokenOR\ tokenEQUAL\ tokenNOTEQUAL\ tokenLESSOREQUAL\ tokenGREATEROREQUAL\ tokenLESS\ tokenGREATER\ tokenPLUS\ tokenMINUS\ tokenMUL\ tokenDIV\ tokenMOD\ tokenGROUPEXPRESSIONBEGIN\ tokenGROUPEXPRESSIONEND\ tokenLRASSIGN\ tokenELSE\ tokenIF\ tokenDO\ tokenFOR\ tokenTO\ tokenDOWNTO\ tokenWHILE\ tokenCONTINUE\ tokenBREAK\	#define TOKENS tokenCOLON\ tokenGOTO\ tokenINTEGER16\ tokenCOMMA\ tokenNOT\ tokenAND\ tokenOR\ tokenEQUAL\ tokenNOTEQUAL\ tokenLESSOREQUAL\ tokenGREATEROREQUAL\ tokenLESS\ tokenGREATER\ tokenPLUS\ tokenMINUS\ tokenMUL\ tokenDIV\ tokenMOD\ tokenGROUPEXPRESSIONBEGIN\ tokenGROUPEXPRESSIONEND\ tokenLRASSIGN\ tokenELSE\ tokenIF\ tokenDO\ tokenFOR\ tokenTO\ tokenDOWNTO\ tokenWHILE\ tokenCONTINUE\ tokenBREAK\	

		"EXIT", "REPEAT", "UNTIL", "GET", "PUT", "NAME", "BODY", "DATA", "BEGIN", "END", "(", ")", "[", "]", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", " ", "~", "A", "B", "C", "D", "E", "F", "G", "H", "I", "J", "K", "L", "M", "N", "O", "P", "Q", "R", "S", "T", "U", "V", "W", "X", "Y", "Z", "a", "b", "c", "d", "e", "f", "g", "h", "i", "j", "k", "l", "m", "n", "o", "p", "q", "r", "s", "t", "u", "v", "w", "x", "y", "z"	"EXIT", "REPEAT", "UNTIL", "GET", "PUT", "NAME", "BODY", "DATA", "BEGIN", "END", "(", ")", "[", "]", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", " ", "~", "A", "B", "C", "D", "E", "F", "G", "H", "I", "J", "K", "L", "M", "N", "O", "P", "Q", "R", "S", "T", "U", "V", "W", "X", "Y", "Z", "a", "b", "c", "d", "e", "f", "g", "h", "i", "j", "k", "l", "m", "n", "o", "p", "q", "r", "s", "t", "u", "v", "w", "x", "y", "z"	tokenEXIT,\ tokenREPEAT,\ tokenUNTIL,\ tokenGET,\ tokenPUT,\ tokenNAME,\ tokenBODY,\ tokenDATA,\ tokenBEGIN,\ tokenEND,\ tokenBEGINBLOCK,\ tokenENDBLOCK,\ tokenLEFTSQUAREBRACKETS,\ tokenRIGHTSQUAREBRACKETS,\ tokenSEMICOLON,\ difit_0,\ difit_1,\ difit_2,\ difit_3,\ difit_4,\ difit_5,\ difit_6,\ difit_7,\ difit_8,\ difit_9,\ tokenUNDERSCORE,\ A,\ B,\ C,\ D,\ E,\ F,\ G,\ H,\ I,\ J,\ K,\ L,\ M,\ N,\ O,\ P,\ Q,\ R,\ S,\ T,\ U,\ V,\ W,\ X,\ Y,\ Z,\ a,\ b,\ c,\ d,\ e,\ f,\ g,\ h,\ i,\ j,\ k,\ l,\ m,\ n,\ o,\ p,\ q,\ r,\ s,\ t,\ u,\ v,\ w,\ x,\ y,\ z	tokenEXIT,\ tokenREPEAT,\ tokenUNTIL,\ tokenGET,\ tokenPUT,\ tokenNAME,\ tokenBODY,\ tokenDATA,\ tokenBEGIN,\ tokenEND,\ tokenBEGINBLOCK,\ tokenENDBLOCK,\ tokenLEFTSQUAREBRACKETS,\ tokenRIGHTSQUAREBRACKETS,\ tokenSEMICOLON,\ difit_0,\ difit_1,\ difit_2,\ difit_3,\ difit_4,\ difit_5,\ difit_6,\ difit_7,\ difit_8,\ difit_9,\ tokenUNDERSCORE,\ A,\ B,\ C,\ D,\ E,\ F,\ G,\ H,\ I,\ J,\ K,\ L,\ M,\ N,\ O,\ P,\ Q,\ R,\ S,\ T,\ U,\ V,\ W,\ X,\ Y,\ Z,\ a,\ b,\ c,\ d,\ e,\ f,\ g,\ h,\ i,\ j,\ k,\ l,\ m,\ n,\ o,\ p,\ q,\ r,\ s,\ t,\ u,\ v,\ w,\ x,\ y,\ z	#define T_BEGIN GROUPEXPRESSION_0 "(" #define T_BEGIN GROUPEXPRESSION_1 "" #define T_BEGIN GROUPEXPRESSION_2 "" #define T_BEGIN GROUPEXPRESSION_3 "" #define T_END GROUPEXPRESSION_0 ")" #define T_END GROUPEXPRESSION_1 "" #define T_END GROUPEXPRESSION_2 "" #define T_END GROUPEXPRESSION_3 "" #define T_LEFT SQUAREBRACKETS_0 "[" #define T_LEFT SQUAREBRACKETS_1 "" #define T_LEFT SQUAREBRACKETS_2 "" #define T_LEFT SQUAREBRACKETS_3 "" #define T_RIGHT SQUAREBRACKETS_0 "]" #define T_RIGHT SQUAREBRACKETS_1 "" #define T_RIGHT SQUAREBRACKETS_2 "" #define T_RIGHT SQUAREBRACKETS_3 "" #define T_BEGIN_BLOCK_0 "(" #define T_BEGIN_BLOCK_1 "" #define T_BEGIN_BLOCK_2 "" #define T_BEGIN_BLOCK_3 "" #define T_END_BLOCK_0 ")" #define T_END_BLOCK_1 "" #define T_END_BLOCK_2 "" #define T_END_BLOCK_3 "" #define T_SEMICOLON_0 ";" #define T_SEMICOLON_1 "" #define T_SEMICOLON_2 "" #define T_SEMICOLON_3 ""
				tokenGROUPEXPRESSIONBEGIN = "(" >> BOUNDARIES;	tokenGROUPEXPRESSIONBEGIN = "(" >> BOUNDARIES;	
				tokenGROUPEXPRESSIONEND = ")" >> BOUNDARIES;	tokenGROUPEXPRESSIONEND = ")" >> BOUNDARIES;	
				tokenLEFTSQUAREBRACKETS = "[" >> BOUNDARIES;	tokenLEFTSQUAREBRACKETS = "[" >> BOUNDARIES;	
				tokenRIGHTSQUAREBRACKETS = "]" >> BOUNDARIES;	tokenRIGHTSQUAREBRACKETS = "]" >> BOUNDARIES;	
				tokenBEGINBLOCK = "{" >> BOUNDARIES;	tokenBEGINBLOCK = "{" >> BOUNDARIES;	
				tokenENDBLOCK = "}" >> BOUNDARIES;	tokenENDBLOCK = "}" >> BOUNDARIES;	
				tokenSEMICOLON = ";" >> BOUNDARIES;	tokenSEMICOLON = ";" >> BOUNDARIES;	

				tokenCOLON = ":" >> BOUNDARIES;	tokenCOLON = ":" >> BOUNDARIES;	#define T_COLON_0 ":" #define T_COLON_1 "" #define T_COLON_2 "" #define T_COLON_3 ""
				tokenGOTO = "GOTO" >> STRICT_BOUNDARIES;	tokenGOTO = "GOTO" >> STRICT_BOUNDARIES;	#define T_GOTO_0 "GOTO" #define T_GOTO_1 "" #define T_GOTO_2 "" #define T_GOTO_3 ""
				tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	#define T_DATA_TYPE_0 "INTEGER16" #define T_DATA_TYPE_1 "" #define T_DATA_TYPE_2 "" #define T_DATA_TYPE_3 ""
				tokenCOMMA = "," >> BOUNDARIES;	tokenCOMMA = "," >> BOUNDARIES;	#define T_COMA_0 "," #define T_COMA_1 "" #define T_COMA_2 "" #define T_COMA_3 ""
						#define T_BITWISE_NOT_0 "~" #define T_BITWISE_NOT_1 "" #define T_BITWISE_NOT_2 "" #define T_BITWISE_NOT_3 ""
				tokenNOT = "NOT" >> STRICT_BOUNDARIES;	tokenNOT = "NOT" >> STRICT_BOUNDARIES;	#define T_NOT_0 "NOT" #define T_NOT_1 "" #define T_NOT_2 "" #define T_NOT_3 ""
						#define T_BITWISE_AND_0 "&" #define T_BITWISE_AND_1 "" #define T_BITWISE_AND_2 "" #define T_BITWISE_AND_3 ""
				tokenAND = "AND" >> STRICT_BOUNDARIES;	tokenAND = "AND" >> STRICT_BOUNDARIES;	#define T_AND_0 "AND" #define T_AND_1 "" #define T_AND_2 "" #define T_AND_3 ""
						#define T_BITWISE_OR_0 " " #define T_BITWISE_OR_1 "" #define T_BITWISE_OR_2 "" #define T_BITWISE_OR_3 ""
				tokenOR = "OR" >> STRICT_BOUNDARIES;	tokenOR = "OR" >> STRICT_BOUNDARIES;	#define T_OR_0 "OR" #define T_OR_1 "" #define T_OR_2 "" #define T_OR_3 ""
				tokenEQUAL = "==" >> BOUNDARIES;	tokenEQUAL = "==" >> BOUNDARIES;	#define T_EQUAL_0 "==" #define T_EQUAL_1 "" #define T_EQUAL_2 "" #define T_EQUAL_3 ""
				tokenNOTEQUAL = "!=" >> BOUNDARIES;	tokenNOTEQUAL = "!=" >> BOUNDARIES;	#define T_NOT_EQUAL_0 "!=" #define T_NOT_EQUAL_1 "" #define T_NOT_EQUAL_2 "" #define T_NOT_EQUAL_3 ""
				tokenLESSOREQUAL = "<=" >> BOUNDARIES;	tokenLESSOREQUAL = "<=" >> BOUNDARIES;	#define T_LESS_OR_EQUAL_0 "<="
						#define T_LESS_OR_EQUAL_1 "" #define T_LESS_OR_EQUAL_2 "" #define T_LESS_OR_EQUAL_3 ""
				tokenGREATEROREQUAL = ">=" >> BOUNDARIES;	tokenGREATEROREQUAL = ">=" >> BOUNDARIES;	#define T_GREATER_OR_EQUAL_0 ">="
						#define T_GREATER_OR_EQUAL_1 "" #define T_GREATER_OR_EQUAL_2 "" #define T_GREATER_OR_EQUAL_3 ""
				tokenLESS = "<" >> BOUNDARIES;	tokenLESS = "<" >> BOUNDARIES;	#define T_LESS_0 "<" #define T_LESS_1 "" #define T_LESS_2 "" #define T_LESS_3 ""
				tokenGREATER = ">" >> BOUNDARIES;	tokenGREATER = ">" >> BOUNDARIES;	#define T_GREATER_0 ">" #define T_GREATER_1 "" #define T_GREATER_2 "" #define T_GREATER_3 ""
				tokenPLUS = "+" >> BOUNDARIES;	tokenPLUS = "+" >> BOUNDARIES;	#define T_ADD_0 "+" #define T_ADD_1 "" #define T_ADD_2 "" #define T_ADD_3 ""
				tokenMINUS = "-" >> BOUNDARIES;	tokenMINUS = "-" >> BOUNDARIES;	#define T_SUB_0 "-" #define T_SUB_1 "" #define T_SUB_2 "" #define T_SUB_3 ""
				tokenMUL = "*" >> BOUNDARIES;	tokenMUL = "*" >> BOUNDARIES;	#define T_MUL_0 "*" #define T_MUL_1 "" #define T_MUL_2 "" #define T_MUL_3 ""
				tokenDIV = "DIV" >> STRICT_BOUNDARIES;	tokenDIV = "DIV" >> STRICT_BOUNDARIES;	#define T_DIV_0 "DIV" #define T_DIV_1 "" #define T_DIV_2 "" #define T_DIV_3 ""
				tokenMOD = "MOD" >> STRICT_BOUNDARIES;	tokenMOD = "MOD" >> STRICT_BOUNDARIES;	#define T_MOD_0 "MOD" #define T_MOD_1 "" #define T_MOD_2 "" #define T_MOD_3 ""
				tokenLRASSIGN = "=:;" >> BOUNDARIES;	tokenLRASSIGN = "=:;" >> BOUNDARIES;	#define T_LRASSIGN_0 "=:;" #define T_LRASSIGN_1 "" #define T_LRASSIGN_2 "" #define T_LRASSIGN_3 ""
						#define T_THEN_BLOCK_0 "{" #define T_THEN_BLOCK_1 "" #define T_THEN_BLOCK_2 "" #define T_THEN_BLOCK_3 ""
				tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	#define T_ELSE_BLOCK_0 "ELSE" #define T_ELSE_BLOCK_1 T_BEGIN_BLOCK_0 #define T_ELSE_BLOCK_2 "" #define T_ELSE_BLOCK_3 ""
				tokenIF = "IF" >> STRICT_BOUNDARIES;	tokenIF = "IF" >> STRICT_BOUNDARIES;	#define T_IF_0 "IF" #define T_IF_1 "" #define T_IF_2 "" #define T_IF_3 ""
						#define T_ELSE_IF_0 "ELSE" #define T_ELSE_IF_1 T_IF_0 #define T_ELSE_IF_2 "" #define T_ELSE_IF_3 ""
				tokenDO = "DO" >> STRICT_BOUNDARIES;	tokenDO = "DO" >> STRICT_BOUNDARIES;	#define T_DO_0 "DO" #define T_DO_1 "" #define T_DO_2 ""

						#define T_DO_3 ""
				tokenFOR = "FOR" >> STRICT_BOUNDARIES;	tokenFOR = "FOR" >> STRICT_BOUNDARIES;	#define T_FOR_0 "FOR" #define T_FOR_1 "" #define T_FOR_2 "" #define T_FOR_3 ""
				tokenTO = "TO" >> STRICT_BOUNDARIES;	tokenTO = "TO" >> STRICT_BOUNDARIES;	#define T_TO_0 "TO" #define T_TO_1 "" #define T_TO_2 "" #define T_TO_3 ""
				tokenDOWNT0 = "DOWNT0" >> STRICT_BOUNDARIES;	tokenDOWNT0 = "DOWNT0" >> STRICT_BOUNDARIES;	#define T_DOWNT0_0 "DOWNT0" #define T_DOWNT0_1 "" #define T_DOWNT0_2 "" #define T_DOWNT0_3 ""
				tokenWHILE = "WHILE" >> STRICT_BOUNDARIES;	tokenWHILE = "WHILE" >> STRICT_BOUNDARIES;	#define T_WHILE_0 "WHILE" #define T_WHILE_1 "" #define T_WHILE_2 "" #define T_WHILE_3 ""
				tokenCONTINUE = "CONTINUE" >> STRICT_BOUNDARIES;	tokenCONTINUE = "CONTINUE" >> STRICT_BOUNDARIES;	#define T_CONTINUE_WHILE_0 "CONTINUE" #define T_CONTINUE_WHILE_1 "" #define T_CONTINUE_WHILE_2 "" #define T_CONTINUE_WHILE_3 ""
				tokenBREAK = "BREAK" >> STRICT_BOUNDARIES;	tokenBREAK = "BREAK" >> STRICT_BOUNDARIES;	#define T_EXIT_WHILE_0 "BREAK" #define T_EXIT_WHILE_1 "" #define T_EXIT_WHILE_2 "" #define T_EXIT_WHILE_3 ""
				tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	#define T_EXIT_0 "EXIT" #define T_EXIT_1 "" #define T_EXIT_2 "" #define T_EXIT_3 ""
				tokenREPEAT = "REPEAT" >> STRICT_BOUNDARIES;	tokenREPEAT = "REPEAT" >> STRICT_BOUNDARIES;	#define T_REPEAT_0 "REPEAT" #define T_REPEAT_1 "" #define T_REPEAT_2 "" #define T_REPEAT_3 ""
				tokenUNTIL = "UNTIL" >> STRICT_BOUNDARIES;	tokenUNTIL = "UNTIL" >> STRICT_BOUNDARIES;	#define T_UNTIL_0 "UNTIL" #define T_UNTIL_1 "" #define T_UNTIL_2 "" #define T_UNTIL_3 ""
				tokenGET = "GET" >> STRICT_BOUNDARIES;	tokenGET = "GET" >> STRICT_BOUNDARIES;	#define T_INPUT_0 "GET" #define T_INPUT_1 "" #define T_INPUT_2 "" #define T_INPUT_3 ""
				tokenPUT = "PUT" >> STRICT_BOUNDARIES;	tokenPUT = "PUT" >> STRICT_BOUNDARIES;	#define T_OUTPUT_0 "PUT" #define T_OUTPUT_1 "" #define T_OUTPUT_2 "" #define T_OUTPUT_3 ""
				tokenNAME = "NAME" >> STRICT_BOUNDARIES;	tokenNAME = "NAME" >> STRICT_BOUNDARIES;	#define T_NAME_0 "NAME" #define T_NAME_1 "" #define T_NAME_2 "" #define T_NAME_3 ""
				tokenBODY = "BODY" >> STRICT_BOUNDARIES;	tokenBODY = "BODY" >> STRICT_BOUNDARIES;	#define T_BODY_0 "BODY" #define T_BODY_1 "" #define T_BODY_2 "" #define T_BODY_3 ""
				tokenDATA = "DATA" >> STRICT_BOUNDARIES;	tokenDATA = "DATA" >> STRICT_BOUNDARIES;	#define T_DATA_0 "DATA" #define T_DATA_1 "" #define T_DATA_2 "" #define T_DATA_3 ""
				tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	#define T_BEGIN_0 "BEGIN" #define T_BEGIN_1 "" #define T_BEGIN_2 "" #define T_BEGIN_3 ""
				tokenEND = "END" >> STRICT_BOUNDARIES;	tokenEND = "END" >> STRICT_BOUNDARIES;	#define T_END_0 "END" #define T_END_1 "" #define T_END_2 "" #define T_END_3 ""
						#define T_NULL_STATEMENT_0 "NULL" #define T_NULL_STATEMENT_1 "STATEMENT" #define T_NULL_STATEMENT_2 "" #define T_NULL_STATEMENT_3 ""
						#define GRAMMAR_LL2_2025 {\
labeled_point = ident , ":";	labeled_point = ident , ":";	labeled_point → ident ":"	labeled_point(1: "ident_terminal") → ident ":"	labeled_point = ident >> tokenCOLON;	labeled_point = ident >> tokenCOLON;	{ LA_IS, {"ident_terminal"}, {"labeled_point"}, {\ LA_IS, {""}, 2, {"ident", T_COLON_0}}\}};\
goto_label = "GOTO" , ident;	goto_label = "GOTO" , ident;	goto_label → "GOTO" ident	goto_label(1: "GOTO") → "GOTO" ident	goto_label = tokenGOTO >> ident;	goto_label = tokenGOTO >> ident;	{ LA_IS, {T_GOTO_0}, {"goto_label"}, {\ LA_IS, {""}, 2, {T_GOTO_0, "ident"}}\}};\
program_name = ident;	program_name = ident;	program_name → ident	program_name(1: "ident_terminal") → ident	program_name = SAME_RULE(ident);	program_name = SAME_RULE(ident);	{ LA_IS, {"ident_terminal"}, {"program_name"}, {\ LA_IS, {""}, 1, {"ident"}}\}};\
value_type = "INTEGER16";	value_type = "INTEGER16";	value_type → "INTEGER16"	value_type(1: "INTEGER16") → "INTEGER16"	value_type = SAME_RULE(tokenINTEGER16);	value_type = SAME_RULE(tokenINTEGER16);	{ LA_IS, {T_DATA_TYPE_0}, {"value_type"}, {\ LA_IS, {""}, 1, {T_DATA_TYPE_0}}\}};\
array_specify = "[" , unsigned_value , "]" ;	array_specify = "[" , unsigned_value , "]" ;	array_specify → "[" unsigned_value "]"	array_specify(1: "["] → "[" unsigned_value "]"		array_specify = "[" >> unsigned_value >> "]" ;	{ LA_IS, {"["], {"array_specify"}, {\ LA_IS, {""}, 3, {"[" , "unsigned_value" , "]"}}\}};\
declaration_element = ident , array_specify__optional ;	declaration_element → ident array_specify__optional	declaration_element(1: "ident_terminal") → ident array_specify__optional	declaration_element(1: "ident_terminal") → ident array_specify__optional	declaration_element = ident >> -(tokenLEFTSQUAREBRACKETS >> unsigned_value >> tokenRIGHTSQUAREBRACKETS);	declaration_element = ident >> array_specify__optional;	{ LA_IS, {"ident_terminal"}, {"declaration_element"}, {\ LA_IS, {""}, 2, {"ident", "array_specify__optional"}}\}};\
array_specify__optional = array_specify ε ;	array_specify__optional → array_specify array_specify__optional → ε	array_specify__optional(1: "["] → array_specify array_specify__optional(1: ! "["] → ε	array_specify__optional(1: "["] → array_specify array_specify__optional(1: ! "["] → ε		array_specify__optional = array_specify "" ;	{ LA_IS, {"["], {"array_specify__optional"}, {\ LA_IS, {""}, 1, {"array_specify"}}\}};\
other_declaration_ident = " , , declaration_element ;	other_declaration_ident → " , , declaration_element	other_declaration_ident(1: " , , ") → " , , declaration_element	other_declaration_ident(1: " , , ") → " , , declaration_element	other_declaration_ident = tokenCOMMA >> declaration_element;	other_declaration_ident = tokenCOMMA >> declaration_element;	{ LA_IS, {T_COMA_0}, {"other_declaration_ident"}, {\ LA_IS, {""}, 2, {T_COMA_0, "declaration_element"}}\}};\
declaration = value_type , declaration_element , other_declaration_ident__iteration ;	declaration → value_type declaration_element other_declaration_ident__iteration	declaration(1: "INTEGER16") → value_type declaration_element other_declaration_ident__iteration	declaration(1: "INTEGER16") → value_type declaration_element other_declaration_ident__iteration	declaration = value_type >> declaration_element >> *other_declaration_ident;	declaration = value_type >> declaration_element >> other_declaration_ident__iteration;	{ LA_IS, {T_DATA_TYPE_0}, {"declaration"}, {\ LA_IS, {""}, 3, {"value_type", "declaration_element", "other_declaration_ident__iteration"}}\}};\
other_declaration_ident__iteration = other_declaration_ident , other_declaration_ident__iteration ε ;	other_declaration_ident__iteration → other_declaration_ident other_declaration_ident__iteration false_cond_block_without_else__iteration → ε	other_declaration_ident__iteration(1: " , , ") → other_declaration_ident other_declaration_ident__iteration(1: ! " , , ") → ε	other_declaration_ident__iteration(1: " , , ") → other_declaration_ident other_declaration_ident__iteration(1: ! " , , ") → ε		other_declaration_ident__iteration = other_declaration_ident >> other_declaration_ident__iteration "" ;	{ LA_IS, {T_COMA_0}, {"other_declaration_ident__iteration"}, {\ LA_IS, {""}, 2, {"other_declaration_ident", "other_declaration_ident__iteration"}}\}};\

						{ LA_NOT, { T_COMA_0 }, { "other_declaration_ident__iteration", {\n { LA_IS, { "" }, 0, { "" } } } \n }} \n
index_action = "[" , expression , "]" ;	index_action = "[" , expression , "]" ;	index_action → "[" expression "]"	index_action(1: "[") → "[" expression "]"	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;	{ LA_IS, { "[" }, { "index_action", {\n { LA_IS, { "" }, 3, { "[", "expression", "]" } } } \n }} \n
unary_operator = "NOT";	unary_operator = "NOT";	unary_operator → "NOT"	unary_operator(1: "NOT") → "NOT"	unary_operator = SAME_RULE(tokenNOT);	unary_operator = SAME_RULE(tokenNOT);	{ LA_IS, { T_NOT_0 }, { "unary_operator", {\n { LA_IS, { "" }, 1, { T_NOT_0 } } } \n }} \n
unary_operation = unary_operator , expression;	unary_operation = unary_operator , expression;	unary_operation → unary_operator expression	unary_operation(1: "NOT") → unary_operator expression	unary_operation = unary_operator >> expression;	unary_operation = unary_operator >> expression;	{ LA_IS, { T_NOT_0 }, { "unary_operation", {\n { LA_IS, { "" }, 2, { "unary_operator", "expression" } } } \n }} \n
binary_operator = "AND" "OR" "==" "!=" "<=" ">=" "<" ">" "+" "-" "*" "DIV" "MOD";	binary_operator = "AND" "OR" "==" "!=" "<=" ">=" "<" ">" "+" "-" "*" "DIV" "MOD";	binary_operator → "AND" binary_operator → "OR" binary_operator → "==" binary_operator → "!=" binary_operator → "<=" binary_operator → ">=" binary_operator → "<" binary_operator → ">" binary_operator → "+" binary_operator → "-" binary_operator → "*" binary_operator → "DIV" binary_operator → "MOD"	binary_operator(1: "AND") → "AND" binary_operator(1: "OR") → "OR" binary_operator(1: "==") → "==" binary_operator(1: "!=") → "!=" binary_operator(1: "<=") → "<=" binary_operator(1: ">=") → ">=" binary_operator(1: "<") → "<" binary_operator(1: ">") → ">" binary_operator(1: "+") → "+" binary_operator(1: "-") → "-" binary_operator(1: "*") → "*" binary_operator(1: "DIV") → "DIV" binary_operator(1: "MOD") → "MOD"	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenLESS tokenGREATER tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenLESS tokenGREATER tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	{ LA_IS, { T_AND_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_AND_0 } } } \n }} \n { LA_IS, { T_OR_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_OR_0 } } } \n }} \n { LA_IS, { T_EQUAL_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_EQUAL_0 } } } \n }} \n { LA_IS, { T_NOT_EQUAL_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_NOT_EQUAL_0 } } } \n }} \n IF_NONUSE_COMPARE_WITH_EQUAL(\n { LA_IS, { T_LESS_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_LESS_0 } } } \n }} \n { LA_IS, { T_GREATER_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_GREATER_0 } } } \n }} \n) \n IF_USE_COMPARE_WITH_EQUAL(\n { LA_IS, { T_LESS_OR_EQUAL_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_LESS_OR_EQUAL_0 } } } \n }} \n { LA_IS, { T_GREATER_OR_EQUAL_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_GREATER_OR_EQUAL_0 } } } \n }} \n) \n { LA_IS, { T_ADD_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_ADD_0 } } } \n }} \n { LA_IS, { T_SUB_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_SUB_0 } } } \n }} \n { LA_IS, { T_MUL_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_MUL_0 } } } \n }} \n { LA_IS, { T_DIV_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_DIV_0 } } } \n }} \n { LA_IS, { T_MOD_0 }, { "binary_operator", {\n { LA_IS, { "" }, 1, { T_MOD_0 } } } \n }} \n
binary_action = binary_operator , expression;	binary_action = binary_operator , expression;	binary_action → binary_operator expression	binary_action(1: "AND", "OR", "==", "!=", "<=", ">=", "<", ">", "+", "-", "*", "DIV", "MOD") → binary_operator expression	binary_action = binary_operator >> expression;	binary_action = binary_operator >> expression;	IF_NONUSE_COMPARE_WITH_EQUAL(\n { LA_IS, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_0, T_GREATER_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action", {\n { LA_IS, { "" }, 2, { "binary_operator", "expression" } } } \n }} \n) \n IF_USE_COMPARE_WITH_EQUAL(\n { LA_IS, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_OR_EQUAL_0, T_GREATER_OR_EQUAL_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action", {\n { LA_IS, { "" }, 2, { "binary_operator", "expression" } } } \n }} \n) \n
left_expression = group_expression unary_operation ident , [index_action] value cond_block;	left_expression = group_expression unary_operation ident , [index_action] value cond_block;	left_expression → group_expression left_expression → unary_operation left_expression → ident , index_action__optional left_expression → value left_expression → cond_block	left_expression(1: "(") → group_expression left_expression(1: "NOT") → unary_operation left_expression(1: "_") → ident , index_action__optional left_expression(1: "0", "1", "2", "3", "4", "5", "6", "7", "8", "9") → value left_expression(1: "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9") → value left_expression(1: "IF") → cond_block	left_expression = group_expression unary_operation ident >> -index_action value cond_block;	left_expression = group_expression unary_operation ident >> index_action__optional value cond_block;	{ LA_IS, { "(" }, { "left_expression", {\n { LA_IS, { "" }, 1, { "group_expression" } } } \n }} \n { LA_IS, { T_NOT_0 }, { "left_expression", {\n { LA_IS, { "" }, 1, { "unary_operation" } } } \n }} \n { LA_IS, { "ident_terminal" }, { "left_expression", {\n { LA_IS, { "" }, 2, { "ident", "index_action__optional" } } } \n }} \n { LA_IS, { "unsigned_value_terminal" }, { "left_expression", {\n { LA_IS, { "" }, 1, { "value" } } } \n }} \n { LA_IS, { T_ADD_0, T_SUB_0 }, { "left_expression", {\n { LA_IS, { "unsigned_value_terminal" }, 1, { "value" } } } \n /* { LA_NOT, { "unsigned_value_terminal" } }, 1, { "unary_operation" } } */ \n }} \n { LA_IS, { T_IF_0 }, { "left_expression", {\n { LA_IS, { "" }, 1, { "cond_block" } } } \n }} \n
	index_action__optional = index_action ε;	index_action__optional → index_action index_action__optional → ε	index_action__optional(1: "(") → index_action index_action__optional(1: !"(") → ε		index_action__optional = index_action_action "";	{ LA_IS, { "(" }, { "index_action__optional", {\n { LA_IS, { "" }, 1, { "index_action" } } } \n }} \n { LA_NOT, { "(" }, { "index_action__optional", {\n { LA_IS, { "" }, 0, { "" } } } \n }} \n
expression = left_expression , binary_action__iteration;	expression = left_expression , binary_action__iteration;	expression → left_expression binary_action__iteration	expression(1: "(" , "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → left_expression binary_action__iteration	expression = left_expression >> *binary_action;	expression = left_expression >> binary_action__iteration;	{ LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression", {\n { LA_IS, { "" }, 2, { "left_expression", "binary_action__iteration" } } } \n }} \n
expression = left_expression , {binary_action};	binary_action__iteration = binary_action, binary_action__iteration ε;	binary_action__iteration → binary_action binary_action__iteration	binary_action__iteration(1: "AND", "OR", "==", "!=", "<=", ">=", "<", ">", "+", "-", "*", "DIV", "MOD") →		binary_action__iteration = binary_action >> binary_action__iteration "";	IF_NONUSE_COMPARE_WITH_EQUAL(\n { LA_IS, { T_AND_0, T_OR_0, T_EQUAL_0,

		binary_action__iteration → ε	binary_action binary_action__iteration binary_action__iteration(1: !"AND", !"OR", !"=", !"!=", !"<=", !">=", !"<", !">", !"+", !"-", !"*", !"DIV", !"MOD") → ε			T_NOT_EQUAL_0, T_LESS_0, T_GREATER_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action__iteration",{\ {LA_IS, !""}, 2, { "binary_action", "binary_action__iteration" }}\ }}\\ {LA_NOT, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_0, T_GREATER_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action__iteration",{\ {LA_IS, !""}, 0, { "" }}\ }}\\ }\\ IF_USE_COMPARE_WITH_EQUAL(\ {LA_IS, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_OR_EQUAL_0, T_GREATER_OR_EQUAL_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action__iteration",{\ {LA_IS, !""}, 2, { "binary_action", "binary_action__iteration" }}\ }}\\ {LA_NOT, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_OR_EQUAL_0, T_GREATER_OR_EQUAL_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action__iteration",{\ {LA_IS, !""}, 0, { "" }}\ }}\\ }\
group_expression = "(" , expression , ")";	group_expression = "(" , expression , ")";	group_expression → "(" expression ")"	group_expression(1: "(") → "(" expression ")"	group_expression = tokenGROUPEXPRESSIONBEGIN >> expression >> tokenGROUPEXPRESSIONEND;	group_expression = tokenGROUPEXPRESSIONBEGIN >> expression >> tokenGROUPEXPRESSIONEND;	{LA_IS, { "(" }, { "group_expression", {\ {LA_IS, !""}, 3, { "(", "expression", ")" }}\ }}\
expression_or_cond_block__with_optional_assign = expression , assign_to_right__optional;	expression_or_cond_block__with_optional_assign = expression , assign_to_right__optional;	expression_or_cond_block__with_optional_assign → expression assign_to_right__optional	expression_or_cond_block__with_optional_assign(1: "(", "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression assign_to_right__optional	expression_or_cond_block__with_optional_assign = expression >> -(tokenLRASSIGN >> ident >> -index_action);	expression_or_cond_block__with_optional_assign = expression >> assign_to_right__optional;	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression_or_cond_block__with_optional_assign", {\ {LA_IS, !""}, 2, { "expression", "assign_to_right__optional" }}\ }}\
expression_or_cond_block__with_optional_assign = expression , ["=" , ident , index_action];	assign_to_right = "=", ident , index_action__optional;	assign_to_right → "=" ident index_action__optional	assign_to_right(1: "=") → "=" ident index_action__optional		assign_to_right = tokenLRASSIGN >> ident >> index_action__optional;	{LA_IS, { T_LRASSIGN_0 }, { "assign_to_right", {\ {LA_IS, !""}, 3, { T_LRASSIGN_0, "ident", "index_action__optional" }}\ }}\
	assign_to_right__optional = assign_to_right ε;	assign_to_right__optional → assign_to_right assign_to_right__optional → ε;	assign_to_right__optional(1: "=") → assign_to_right assign_to_right__optional(1: !"=") → ε;		assign_to_right__optional = assign_to_right "";	{ LA_IS, { T_LRASSIGN_0 }, { "assign_to_right__optional", {\ { LA_IS, !""}, 1, { "assign_to_right" }}\ }}\\ {LA_NOT, { T_LRASSIGN_0 }, { "assign_to_right__optional", {\ { LA_IS, !""}, 0, { "" }}\ }}\
if_expression = expression;	if_expression = expression;	if_expression → expression	if_expression(1: "(", "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	if_expression = SAME_RULE(expression);	if_expression = SAME_RULE(expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "if_expression", {\ {LA_IS, !""}, 1, { "expression" }}\ }}\
body_for_true = block_statements_in_while_and_if_body;	body_for_true = block_statements_in_while_and_if_body;	body_for_true → block_statements_in_while_and_if_body	body_for_true(1: "{") → block_statements_in_while_and_if_body	body_for_true = SAME_RULE(block_statements_in_while_and_if_body);	body_for_true = SAME_RULE(block_statements_in_while_and_if_body);	{LA_IS, { T_BEGIN_BLOCK_0 }, { "body_for_true", {\ {LA_IS, !""}, 1, { "block_statements_in_while_and_if_body" }}\ }}\
false_cond_block_without_else = "ELSE" , "IF" , if_expression , body_for_true;	false_cond_block_without_else = "ELSE" , "IF" , if_expression , body_for_true;	false_cond_block_without_else → "ELSE" "IF" if_expression body_for_true	false_cond_block_without_else(1: "ELSE") → "ELSE" "IF" if_expression body_for_true	false_cond_block_without_else = tokenELSE >> tokenIF >> if_expression >> body_for_true;	false_cond_block_without_else = tokenELSE >> tokenIF >> if_expression >> body_for_true;	{LA_IS, { T_ELSE_IF_0 }, { "false_cond_block_without_else", {\ {LA_IS, !""}, 4, { T_ELSE_IF_0, T_ELSE_IF_1, "if_expression", "body_for_true" }}\ }}\
body_for_false = "ELSE" , block_statements_in_while_and_if_body;	body_for_false = "ELSE", block_statements_in_while_and_if_body;	body_for_false → "ELSE" block_statements_in_while_and_if_body	body_for_false(1: "ELSE") → "ELSE" block_statements_in_while_and_if_body	body_for_false = tokenELSE >> block_statements_in_while_and_if_body;	body_for_false = tokenELSE >> block_statements_in_while_and_if_body;	{LA_IS, { T_ELSE_BLOCK_0 }, { "body_for_false", {\ {LA_IS, !""}, 2, { T_ELSE_BLOCK_0, "block_statements" }}\ }}\
cond_block = "IF" , if_expression , body_for_true, { false_cond_block_without_else } , body_for_false;	cond_block = "IF" , if_expression , body_for_true, false_cond_block_without_else__iteration , body_for_false__optional;	cond_block → "IF" if_expression body_for_true false_cond_block_without_else__iteration body_for_false__optional	cond_block(1: "IF") → "IF" if_expression body_for_true false_cond_block_without_else__iteration body_for_false__optional	cond_block = tokenIF >> if_expression >> body_for_true >> *false_cond_block_without_else >> -body_for_false;	cond_block = tokenIF >> if_expression >> body_for_true >> false_cond_block_without_else__iteration >> body_for_false__optional;	{LA_IS, { T_IF_0 }, { "cond_block", {\ {LA_IS, !""}, 5, { T_IF_0, "if_expression", "body_for_true", "false_cond_block_without_else__iteration", "body_for_false__optional" }}\ }}\
false_cond_block_without_else__iteration = false_cond_block_without_else, false_cond_block_without_else__iteration ε;	false_cond_block_without_else__iteration = false_cond_block_without_else, false_cond_block_without_else__iteration ε;	false_cond_block_without_else__iteration → false_cond_block_without_else false_cond_block_without_else__iteration false_cond_block_without_else__iteration → ε	false_cond_block_without_else__iteration(1: "ELSE"; 2: "IF") → false_cond_block_without_else false_cond_block_without_else__iteration false_cond_block_without_else__iteration(1: "ELSE"; 2: !"IF") → ε false_cond_block_without_else__iteration(1: !"ELSE") → ε	false_cond_block_without_else__iteration = false_cond_block_without_else, false_cond_block_without_else__iteration "";	false_cond_block_without_else__iteration = false_cond_block_without_else, false_cond_block_without_else__iteration "";	{LA_IS, { T_ELSE_IF_0 }, { "false_cond_block_without_else__iteration", {\ {LA_IS, { T_ELSE_IF_1 }, 2, { "false_cond_block_without_else", "false_cond_block_without_else__iteration" }}\, {LA_NOT, { T_ELSE_IF_1 }, 0, { "" }}\ }}\\ {LA_NOT, { T_ELSE_IF_0 }, { "false_cond_block_without_else__iteration", {\ {LA_IS, !""}, 0, { "" }}\ }}\
body_for_false__optional = body_for_false ε;	body_for_false__optional = body_for_false ε;	body_for_false__optional → body_for_false body_for_false__optional → ε	body_for_false__optional(1: "FALSE") → body_for_false body_for_false__optional(1: !"FALSE") → ε		body_for_false__optional = body_for_false "";	{LA_IS, { T_ELSE_BLOCK_0 }, { "body_for_false__optional", {\ {LA_IS, !""}, 1, { "body_for_false" }}\ }}\\ {LA_NOT, { T_ELSE_BLOCK_0 }, { "body_for_false__optional", {\ {LA_IS, !""}, 0, { "" }}\ }}\
cycle_begin_expression = expression;	cycle_begin_expression = expression;	cycle_begin_expression → expression	cycle_begin_expression(1: "(", "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	cycle_begin_expression = SAME_RULE(expression);	cycle_begin_expression = SAME_RULE(expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_begin_expression", {\ {LA_IS, !""}, 1, { "expression" }}\ }}\
cycle_end_expression = expression;	cycle_end_expression = expression;	cycle_end_expression → expression	cycle_end_expression(1: "(", "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	cycle_end_expression = SAME_RULE(expression);	cycle_end_expression = SAME_RULE(expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_end_expression", {\ {LA_IS, !""}, 1, { "expression" }}\ }}\
cycle_counter = ident;	cycle_counter = ident;	cycle_counter → ident	cycle_counter(1: " _") → ident	cycle_counter = SAME_RULE(ident);	cycle_counter = SAME_RULE(ident);	{LA_IS, { "ident_terminal" }, { "cycle_counter", {\ {LA_IS, !""}, 1, { "ident" }}\ }}\

cycle_counter_lr_init = cycle_begin_expression , "=" , cycle_counter;	cycle_counter_lr_init = cycle_begin_expression , "=" , cycle_counter;	cycle_counter_lr_init → cycle_begin_expression "=" cycle_counter	cycle_counter_lr_init(1: "(" , "NOT" , "+" , "-" , "_" , "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , "IF") → cycle_begin_expression "=" cycle_counter	cycle_counter_lr_init = cycle_begin_expression >> tokenLRASSIGN >> cycle_counter;	cycle_counter_lr_init = cycle_begin_expression >> tokenLRASSIGN >> cycle_counter;	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_counter_lr_init", {\n {LA_IS, {""}, 3, { "cycle_begin_expression", T_LRASSIGN_0, "cycle_counter" }}\n }}}\n
cycle_counter_init = cycle_counter_lr_init;	cycle_counter_init = cycle_counter_lr_init;	cycle_counter_init → cycle_counter_lr_init	cycle_counter_init(1: "(" , "NOT" , "+" , "-" , "_" , "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , "IF") → cycle_counter_lr_init	cycle_counter_init = SAME_RULE(cycle_counter_lr_init);	cycle_counter_init = SAME_RULE(cycle_counter_lr_init);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_counter_init", {\n {LA_IS, {""}, 1, { "cycle_counter_lr_init" }}\n }}}\n
cycle_counter_last_value = cycle_end_expression;	cycle_counter_last_value = cycle_end_expression;	cycle_counter_last_value → cycle_end_expression	cycle_counter_last_value(1: "(" , "NOT" , "+" , "-" , "_" , "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , "IF") → cycle_end_expression	cycle_counter_last_value = SAME_RULE(cycle_end_expression);	cycle_counter_last_value = SAME_RULE(cycle_end_expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_counter_last_value", {\n {LA_IS, {""}, 1, { "cycle_end_expression" }}\n }}}\n
cycle_body = "DO" , statements__or__block_statements;	cycle_body = "DO" , statements__or__block_statements;	cycle_body → "DO" statements__or__block_statements	cycle_body(1: "DO") → "DO" statements__or__block_statements	cycle_body = tokenDO >> (statement block_statements);	cycle_body = tokenDO >> statements__or__block_statements;	{LA_IS, { T_DO_0 }, { "cycle_body", {\n {LA_IS, {""}, 2, { T_DO_0, "statements__or__block_statements" }}\n }}}\n
forto_direction = "TO" "DOWNT0";	forto_direction = "TO" "DOWNT0";	forto_direction → "TO" forto_direction → "DOWNT0"	forto_direction(1: "TO") → "TO" forto_direction(1: "DOWNT0") → "DOWNT0"	forto_direction = tokenDO >> (statement block_statements);	forto_direction = tokenDO tokenDOWNT0;	{LA_IS, { T_TO_0 }, { "forto_direction", {\n {LA_IS, {""}, 1, { T_TO_0 }}\n }}}\n {LA_IS, { T_DOWNT0_0 }, { "forto_direction", {\n {LA_IS, {""}, 1, { T_DOWNT0_0 }}\n }}}\n
forto_cycle = "FOR" , cycle_counter_init , forto_direction, cycle_counter_last_value , cycle_body;	forto_cycle = "FOR" , cycle_counter_init , forto_direction, cycle_counter_last_value , cycle_body;	forto_cycle → "FOR" cycle_counter_init forto_direction, cycle_counter_last_value cycle_body	forto_cycle(1: "FOR") → "FOR" cycle_counter_init forto_direction, cycle_counter_last_value cycle_body	forto_cycle = tokenFOR >> cycle_counter_init >> (tokenTO tokenDOWNT0) >> cycle_counter_last_value >> cycle_body;	forto_cycle = tokenFOR >> cycle_counter_init >> forto_direction >> cycle_counter_last_value >> cycle_body;	{LA_IS, { T_FOR_0 }, { "forto_cycle", {\n {LA_IS, {""}, 5, { T_FOR_0, "cycle_counter_init", "forto_direction", "cycle_counter_last_value", "cycle_body" }}\n }}}\n
continue_while = "CONTINUE";	continue_while = "CONTINUE";	continue_while → "CONTINUE"	continue_while(1: "CONTINUE") → "CONTINUE"	continue_while = SAME_RULE(tokenCONTINUE);	continue_while = SAME_RULE(tokenCONTINUE);	{LA_IS, { T_CONTINUE_WHILE_0 }, { "continue_while", {\n {LA_IS, {""}, 1, { T_CONTINUE_WHILE_0 }}\n }}}\n
break_while = "BREAK";	break_while = "BREAK";	break_while → "BREAK"	break_while(1: "BREAK") → "BREAK"	break_while = SAME_RULE(tokenBREAK);	break_while = SAME_RULE(tokenBREAK);	{LA_IS, { T_EXIT_WHILE_0 }, { "break_while", {\n {LA_IS, {""}, 1, { T_EXIT_WHILE_0 }}\n }}}\n
statement_in_while_and_if_body = statement "CONTINUE" "BREAK";	statement_in_while_and_if_body = statement "CONTINUE" "BREAK";	statement_in_while_and_if_body → statement statement_in_while_and_if_body → continue_while statement_in_while_and_if_body(1: "BREAK") → break_while	statement_in_while_and_if_body(1: "_" , "(" , "NOT", "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , "+", "-", "IF" , "FOR" , "WHILE" , "REPEAT" , "GOTO" , "GET" , "PUT" , ",") → statement statement_in_while_and_if_body(1: "CONTINUE") → continue_while statement_in_while_and_if_body(1: "BREAK") → break_while	statement_in_while_and_if_body = statement continue_while break_while;	statement_in_while_and_if_body = statement continue_while break_while;	{LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_FOR_0, T_WHILE_0, T_REPEAT_0, T_GOTO_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement_in_while_and_if_body", {\n {LA_IS, {""}, 1, { "statement" }}\n }}}\n {LA_IS, { T_CONTINUE_WHILE_0 }, { "statement_in_while_and_if_body", {\n {LA_IS, {""}, 1, { "continue_while" }}\n }}}\n {LA_IS, { T_EXIT_WHILE_0 }, { "statement_in_while_and_if_body", {\n {LA_IS, {""}, 1, { "break_while" }}\n }}}\n
block_statements_in_while_and_if_body = "(" , {statement_in_while_and_if_body} , ")";	block_statements_in_while_and_if_body = "(" , {statement_in_while_and_if_body} , ")";	block_statements_in_while_and_if_body → "(" statement_in_while_and_if_body_iteration ")"	block_statements_in_while_and_if_body(1: "(") → "(" statement_in_while_and_if_body_iteration ")"	block_statements_in_while_and_if_body = tokenBEGINBLOCK >> *statement_in_while_and_if_body >> tokenENDBLOCK;	block_statements_in_while_and_if_body = tokenBEGINBLOCK >> statement_in_while_and_if_body_iteration >> tokenENDBLOCK;	{LA_IS, { T_BEGIN_BLOCK_0 }, { "block_statements_in_while_and_if_body", {\n {LA_IS, {""}, 3, { T_BEGIN_BLOCK_0, "statement_in_while_and_if_body_iteration", T_END_BLOCK_0 }}\n }}}\n
statement_in_while_and_if_body_iteration = statement_in_while_and_if_body , statement_in_while_and_if_body_iteration ε;	statement_in_while_and_if_body_iteration = statement_in_while_and_if_body , statement_in_while_and_if_body_iteration ε;	statement_in_while_and_if_body_iteration → statement_in_while_and_if_body statement_in_while_and_if_body_iteration → ε	statement_in_while_and_if_body_iteration(1: "_" , "(", "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-", "IF", "FOR", "WHILE", "REPEAT", "GOTO", "GET", "PUT", ",", "CONTINUE", ",") → statement_in_while_and_if_body statement_in_while_and_if_body_iteration statement_in_while_and_if_body_iteration(1: "!" , "(", "!"NOT", "!"0", "!"1", "!"2", "!"3", "!"4", "!"5", "!"6", !"7", "!"8", "!"9", "!"+", "!"-", "!"IF", "!"FOR", "!"WHILE", !"REPEAT", "!"GOTO", "!"GET", "!"PUT", "!", ",", !"CONTINUE", "!"BREAK") → ε	statement_in_while_and_if_body_iteration = statement_in_while_and_if_body >> statement_in_while_and_if_body_iteration "";	statement_in_while_and_if_body_iteration = statement_in_while_and_if_body >> statement_in_while_and_if_body_iteration "";	{LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_FOR_0, T_WHILE_0, T_REPEAT_0, T_GOTO_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0, T_CONTINUE_WHILE_0, T_EXIT_WHILE_0 }, {\n "statement_in_while_and_if_body_iteration", {\n {LA_IS, {""}, 2, { "statement_in_while_and_if_body", "statement_in_while_and_if_body_iteration" }}\n }}}\n {LA_NOT, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_FOR_0, T_WHILE_0, T_REPEAT_0, T_GOTO_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0, T_CONTINUE_WHILE_0, T_EXIT_WHILE_0 }, {\n "statement_in_while_and_if_body_iteration", {\n {LA_IS, {""}, 0, { "" }}\n }}}\n
while_cycle_head_expression = expression;	while_cycle_head_expression = expression;	while_cycle_head_expression → expression	while_cycle_head_expression(1: "(" , "NOT" , "+" , "-" , "_" , "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , "IF") → expression	while_cycle_head_expression = SAME_RULE(expression);	while_cycle_head_expression = SAME_RULE(expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "while_cycle_head_expression", {\n {LA_IS, {""}, 1, { "expression" }}\n }}}\n
while_cycle = "WHILE" , while_cycle_head_expression , block_statements_in_while_and_if_body;	while_cycle = "WHILE" , while_cycle_head_expression , block_statements_in_while_and_if_body;	while_cycle → "WHILE" while_cycle_head_expression block_statements_in_while_and_if_body	while_cycle(1: "WHILE") → "WHILE" while_cycle_head_expression block_statements_in_while_and_if_body	while_cycle = tokenWHILE >> while_cycle_head_expression >> block_statements_in_while_and_if_body;	while_cycle = tokenWHILE >> while_cycle_head_expression >> block_statements_in_while_and_if_body;	{LA_IS, { T_WHILE_0 }, { "while_cycle", {\n {LA_IS, {""}, 3, { T_WHILE_0, "while_cycle_head_expression", "block_statements_in_while_and_if_body" }}\n }}}\n
repeat_until_cycle_cond = expression;	repeat_until_cycle_cond = expression;	repeat_until_cycle_cond → expression	repeat_until_cycle_cond(1: "(" , "NOT" , "+" , "-" , "_" , "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , "IF") → expression	repeat_until_cycle_cond = SAME_RULE(expression);	repeat_until_cycle_cond = SAME_RULE(expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "repeat_until_cycle_cond", {\n {LA_IS, {""}, 1, { "expression" }}\n }}}\n
repeat_until_cycle = "REPEAT" , statements__or__block_statements, "UNTIL" , repeat_until_cycle_cond;	repeat_until_cycle = "REPEAT" , statements__or__block_statements, "UNTIL" , repeat_until_cycle_cond;	repeat_until_cycle → "REPEAT" statements__or__block_statements "UNTIL" repeat_until_cycle_cond	repeat_until_cycle(1: "REPEAT") → "REPEAT" statements__or__block_statements "UNTIL" repeat_until_cycle_cond	repeat_until_cycle = tokenREPEAT >> (*statement block_statements) >> tokenUNTIL >> repeat_until_cycle_cond;	repeat_until_cycle = tokenREPEAT >> statements__or__block_statements >> tokenUNTIL >> repeat_until_cycle_cond;	{LA_IS, { T_REPEAT_0 }, { "repeat_until_cycle", {\n {LA_IS, {""}, 4, { T_REPEAT_0, "statements__or__block_statements", T_UNTIL_0, "repeat_until_cycle_cond" }}\n }}}\n
statements__or__block_statements = statement_iteration block_statements;	statements__or__block_statements = statement_iteration block_statements;	statements__or__block_statements → statement_iteration statements__or__block_statements → block_statements	statements__or__block_statements(1: "_" , "(" , "NOT", "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , "+", "-", "IF", "FOR", "WHILE", "REPEAT", "GOTO", "GET", "PUT", ",") → statement_iteration statements__or__block_statements(1: "(") → block_statements	statements__or__block_statements = statement_iteration block_statements;	statements__or__block_statements = statement_iteration block_statements;	{LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_FOR_0, T_WHILE_0, T_REPEAT_0, T_GOTO_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, {\n "statements__or__block_statements", {\n {LA_IS, {""}, 1, { "statement_iteration" }}\n }}}\n {LA_IS, { T_BEGIN_BLOCK_0 }, { "statements__or__block_statements", {\n {LA_IS, {""}, 1, { "block_statements" }}\n }}}\n
input_rule = "GET" , (ident , [index_action] "(" , ident , [index_action] , ")");	input_rule = "GET" , (ident , [index_action] "(" , ident , [index_action] , ")");	input_rule → "GET" argument_for_input	input_rule(1: "GET") → "GET" argument_for_input	input_rule = tokenGET >> (ident >> -index_action tokenGROUPEXPRESSIONBEGIN >> ident	input_rule = tokenGET >> argument_for_input;	{LA_IS, { T_INPUT_0 }, { "input_rule", {\n

				>> -index_action >> tokenGROUPEXPRESSIIONEND);		{LA_IS, {""}, 2, {T_INPUT_0, "argument_for_input"}}\ }}\
	argument_for_input = ident , index_action_optional; argument_for_input = "(" , "ident", "index_action__optional", ")";	argument_for_input → ident index_action_optional argument_for_input → "(" "ident" "index_action__optional" ")"	argument_for_input(1: "-") → ident index_action_optional argument_for_input(1: "(") → "(" "ident" "index_action__optional" ")"		argument_for_input = ident >> index_action__optional tokenGROUPEXPRESSIIONBEGIN >> ident >> index_action_optional >> tokenGROUPEXPRESSIIONEND;	{LA_IS, { "ident_terminal" }, { "argument_for_input", \ {LA_IS, {""}, 2, { "ident", "index_action__optional" }}\ }}\ {LA_IS, { "(" }, { "argument_for_input", \ {LA_IS, {""}, 4, { "(" , "ident", "index_action__optional", ")" }}\ }}\
output = "PUT" , expression;	output = "PUT" , expression;	output → "PUT" expression	output(1: "PUT") → "PUT" expression	output = tokenPUT >> expression;	output = tokenPUT >> expression;	{LA_IS, { T_OUTPUT_0 }, { "output_rule", \ {LA_IS, { "" }, 2, {T_OUTPUT_0, "expression" }}\ }}\
statement = expression_or_cond_block__with_optional_assign labeled_point goto_label forto_cycle while_cycle repeat_until_cycle input_rule output_rule ";"	statement = expression_or_cond_block__with_optional_assign labeled_point goto_label forto_cycle while_cycle repeat_until_cycle input_rule output_rule ";"	statement → expression_or_cond_block__with_optional_assign statement → goto_label statement → labeled_point statement → forto_cycle statement → while_cycle statement → repeat_until_cycle statement → input_rule statement → output_rule statement → ";"	statement(1: "(" , "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression_or_cond_block__with_optional_assign statement(1: "-" , 2: "!") → expression_or_cond_block__with_optional_assign statement(1: "_", ".") → labeled_point statement(1: "GOTO") → goto_label statement(1: "FOR") → forto_cycle statement(1: "WHILE") → while_cycle statement(1: "REPEAT") → repeat_until_cycle statement(1: "GET") → input_rule statement(1: "PUT") → output_rule statement(1: ";") → ";"	statement = expression_or_cond_block__with_optional_assign labeled_point goto_label forto_cycle while_cycle repeat_until_cycle input_rule output_rule tokenSEMICOLON;	statement = expression_or_cond_block__with_optional_assign labeled_point goto_label forto_cycle while_cycle repeat_until_cycle input_rule output_rule tokenSEMICOLON;	{LA_IS, { "(" , T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0 }, { "statement", \ { LA_IS, {""}, 1, ("expression_or_cond_block__with_optional_assign")\ }}\ {LA_IS, { "ident_terminal" }, { "statement", \ { LA_NOT, { T_COLON_0 }, 1, ("expression_or_cond_block__with_optional_assign")\ { LA_IS, { T_COLON_0 }, 1, {"labeled_point" }}\ }}\ {LA_IS, { T_GOTO_0 }, { "statement", \ {LA_IS, {""}, 1, {"goto_label" }}\ }}\ {LA_IS, { T_FOR_0 }, { "statement", \ {LA_IS, {""}, 1, {"forto_cycle" }}\ }}\ {LA_IS, { T_WHILE_0 }, { "statement", \ {LA_IS, {""}, 1, {"while_cycle" }}\ }}\ {LA_IS, { T_REPEAT_0 }, { "statement", \ {LA_IS, {""}, 1, {"repeat_until_cycle" }}\ }}\ {LA_IS, { T_INPUT_0 }, { "statement", \ {LA_IS, {""}, 1, {"input_rule" }}\ }}\ {LA_IS, { T_OUTPUT_0 }, { "statement", \ {LA_IS, {""}, 1, {"output_rule" }}\ }}\ {LA_IS, { T_SEMICOLON_0 }, { "statement", \ {LA_IS, {""}, 1, {";" }}\ }}\
	statement__iteration = statement, statement__iteration ε;	statement__iteration → statement statement__iteration statement__iteration → ε	statement__iteration(1: "-" , "(" , "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-", "IF", "FOR", "WHILE", "REPEAT", "GOTO", "GET", "PUT", ";") → statement statement__iteration statement__iteration(1: !"_", !"(", !"NOT", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"+", !"-", !"IF", !"FOR", !"WHILE", !"REPEAT", !"GOTO", !"GET", !"PUT", !";") → ε		statement__iteration = statement >> statement__iteration ""	{ LA_IS, { "ident_terminal", "(" , T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_FOR_0, T_WHILE_0, T_REPEAT_0, T_REPEAT_0, T_GOTO_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement__iteration", \ { LA_IS, {""}, 2, { "statement", "statement__iteration" }}\ }}\ { LA_NOT, { "ident_terminal", "(" , T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_FOR_0, T_WHILE_0, T_REPEAT_0, T_REPEAT_0, T_GOTO_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement__iteration", \ { LA_IS, {""}, 0, { "" }}\ }}\
block_statements = "(" , {statement} , ")"	block_statements = "(" , {statement} , ")"	block_statements → "(" statement__iteration ")"	block_statements(1: "(") → "(" statement__iteration ")"	block_statements = tokenBEGINBLOCK >> *statement >> tokenENDBLOCK;	block_statements = tokenBEGINBLOCK >> statement__iteration >> tokenENDBLOCK;	{ LA_IS, { T_BEGIN_BLOCK_0 }, { "block_statements", \ { LA_IS, {""}, 3, {T_BEGIN_BLOCK_0, "statement__iteration", T_END_BLOCK_0 }}\ }}\
	expression__optional = expression ""	expression__optional → expression expression__optional → ε	expression__optional(1: "(" , "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression expression__optional(1: !"(", !"NOT", !"+", !"-", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"IF") → ε		expression__optional = expression ""	{LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression__optional", \ {LA_IS, {""}, 1, { "expression" }}\ }}\ {LA_NOT, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression__optional", \ {LA_IS, {""}, 0, { "" }}\ }}\
program_rule = "NAME" , program_name , "+", "BODY", "DATA", declaration__optional , "+", statement__iteration , "END";	program_rule = "NAME" , program_name , "+", "BODY", "DATA", declaration__optional , "+", statement__iteration "END"	program_rule → "NAME" program_name "+", "BODY" "DATA" declaration__optional "+", statement__iteration "END"	program_rule(1: "NAME") → "NAME" program_name "+", "BODY" "DATA" declaration__optional "+", statement__iteration "END"	program_rule = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenBODY >> tokenDATA >> (-declaration) >> tokenSEMICOLON >> *statement >> tokenEND;	program_rule = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenDATA >> declaration__optional >> tokenSEMICOLON >> statement__iteration >> tokenEND;	{ LA_IS, { T_NAME_0 }, { "program_rule", \ { LA_IS, {""}, 9, {T_NAME_0, "program_name", T_SEMICOLON_0, T_BODY_0, T_DATA_0, "declaration__optional", T_SEMICOLON_0, "statement__iteration", T_END_0 }}\ }}\
	declaration__optional = declaration ""	declaration__optional → declaration declaration__optional → ε	declaration__optional(1: "INTEGER16") → declaration declaration__optional(1: !"INTEGER16") → ε		declaration__optional = declaration ""	{ LA_IS, { T_DATA_TYPE_0 }, { "declaration__optional", \ { LA_IS, {""}, 1, { "declaration" }}\ }}\ { LA_NOT, { T_DATA_TYPE_0 }, { "declaration__optional", \ { LA_IS, {""}, 0, { "" }}\ }}\
value = [sign] , unsigned_value;	value = sign__optional, unsigned_value;	value → sign__optional unsigned_value	value(1: "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-") → sign__optional unsigned_value	value = -sign >> unsigned_value >> BOUNDARIES;	value = sign__optional >> unsigned_value >> BOUNDARIES;	{LA_IS, { "unsigned_value_terminal", T_ADD_0, T_SUB_0 }, { "value", \ {LA_IS, { "" }, 2, { "sign__optional", "unsigned_value" }}\ }}\
	sign__optional = sign ε;	sign__optional → sign sign__optional → ε	sign__optional(1: "+" , "-") → sign sign__optional(1: !"+", !"-") → ε		sign__optional = sign ""	{LA_IS, { T_ADD_0, T_SUB_0 }, { "sign__optional", \ {LA_IS, { "" }, 1, { "sign" }}\ }}\ {LA_NOT, { T_ADD_0, T_SUB_0 }, { "sign__optional", \ {LA_IS, { "" }, 0, { "" }}\ }}\
sign = sign_plus sign_minus;	sign → sign_plus sign → sign_minus	sign → sign_plus sign → sign_minus	sign(1: "+") → sign_plus sign(1: "-") → sign_minus	sign = sign_plus sign_minus;	sign = sign_plus sign_minus;	{LA_IS, { T_ADD_0 }, { "sign", \ {LA_IS, {""}, 1, { "sign_plus" }}\ }}\ {LA_IS, { T_SUB_0 }, { "sign", \ {LA_IS, {""}, 1, { "sign_minus" } } \ }}\
sign_plus = "+";	sign_plus = "+";	sign_plus → "+"	sign_plus(1: "+") → "+"	sign_plus = SAME_RULE(tokenPLUS);	sign_plus = SAME_RULE(tokenPLUS);	{LA_IS, { T_ADD_0 }, { "sign_plus", \ {LA_IS, {""}, 1, {T_ADD_0}}\ }}\
sign_minus = "-";	sign_minus = "-";	sign_minus → "-"	sign_minus(1: "-") → "-"	sign_minus = SAME_RULE(tokenMINUS);	sign_minus = SAME_RULE(tokenMINUS);	{LA_IS, { T_SUB_0 }, { "sign_minus", \ {LA_IS, {""}, 1, {T_SUB_0}}\ }}\

unsigned_value = non_zero_digit , {digit} "0";	unsigned_value = non_zero_digit , digit_iteration "0";	unsigned_value → non_zero_digit digit_iteration unsigned_value → "0"	unsigned_value(1: "1", "2", "3", "4", "5", "6", "7", "8", "9") → non_zero_digit digit_iteration unsigned_value(1: "0") → "0"	unsigned_value = (non_zero_digit >> *digit digit_0) >> BOUNDARIES;	unsigned_value = (non_zero_digit >> digit_iteration digit_0) >> BOUNDARIES;	/* unsigned_value_token represents unsigned_value in lexical analyzer */\n{LA_IS, { "unsigned_value_terminal" }, {\n {unsigned_value", {\n {LA_IS, { "" }, 1, { "unsigned_value_terminal" }}\n }}\n}
	digit_iteration = digit, digit_iteration ε;	digit_iteration → digit digit_iteration digit_iteration → ε	digit_iteration(1: "0", "1", "2", "3", "4", "5", "6", "7", "8", "9") → digit digit_iteration digit_iteration(1: !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9") → ε		digit_iteration = digit, digit_iteration "";	\\
digit = "0" non_zero_digit;	digit = "0" non_zero_digit;	digit → "0" digit → non_zero_digit	digit(1: "0") → "0" digit(1: "1", "2", "3", "4", "5", "6", "7", "8", "9") → non_zero_digit	digit_0 = '0'; digit = digit_0 non_zero_digit;	digit_0 = '0'; digit = digit_0 non_zero_digit;	\\
non_zero_digit = "1" "2" "3" "4" "5" "6" "7" "8" "9";	non_zero_digit = "1" "2" "3" "4" "5" "6" "7" "8" "9";	non_zero_digit → "1" non_zero_digit → "2" non_zero_digit → "3" non_zero_digit → "4" non_zero_digit → "5" non_zero_digit → "6" non_zero_digit → "7" non_zero_digit → "8" non_zero_digit → "9"	non_zero_digit(1: "1") → "1" non_zero_digit(1: "2") → "2" non_zero_digit(1: "3") → "3" non_zero_digit(1: "4") → "4" non_zero_digit(1: "5") → "5" non_zero_digit(1: "6") → "6" non_zero_digit(1: "7") → "7" non_zero_digit(1: "8") → "8" non_zero_digit(1: "9") → "9"	digit_1 = '1'; digit_2 = '2'; digit_3 = '3'; digit_4 = '4'; digit_5 = '5'; digit_6 = '6'; digit_7 = '7'; digit_8 = '8'; digit_9 = '9'; non_zero_digit = digit_1 digit_2 digit_3 digit_4 digit_5 digit_6 digit_7 digit_8 digit_9;	digit_1 = '1'; digit_2 = '2'; digit_3 = '3'; digit_4 = '4'; digit_5 = '5'; digit_6 = '6'; digit_7 = '7'; digit_8 = '8'; digit_9 = '9'; non_zero_digit = digit_1 digit_2 digit_3 digit_4 digit_5 digit_6 digit_7 digit_8 digit_9;	\\
ident = "_", letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case ;	ident = "_", letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case ;	ident → "_" letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case	Ident(1: "_") → "_" letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case	tokenUNDERSCORE = "_", ident = tokenUNDERSCORE >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> STRICT_BOUNDARIES;	tokenUNDERSCORE = "_", ident = tokenUNDERSCORE >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> STRICT_BOUNDARIES;	/* ident_token represents ident in lexical analyzer */\n{LA_IS, { "ident_terminal" }, { "ident", {\n {LA_IS, { "" }, 1, { "ident_terminal" }}\n }}\n}
letter_in_lower_case = "a" "b" "c" "d" "e" "f" "g" "h" "i" "j" "k" "l" "m" "n" "o" "p" "q" "r" "s" "t" "u" "v" "w" "x" "y" "z";	letter_in_lower_case = "a" "b" "c" "d" "e" "f" "g" "h" "i" "j" "k" "l" "m" "n" "o" "p" "q" "r" "s" "t" "u" "v" "w" "x" "y" "z";	letter_in_lower_case → "a" letter_in_lower_case → "b" letter_in_lower_case → "c" letter_in_lower_case → "d" letter_in_lower_case → "e" letter_in_lower_case → "f" letter_in_lower_case → "g" letter_in_lower_case → "h" letter_in_lower_case → "i" letter_in_lower_case → "j" letter_in_lower_case → "k" letter_in_lower_case → "l" letter_in_lower_case → "m" letter_in_lower_case → "n" letter_in_lower_case → "o" letter_in_lower_case → "p" letter_in_lower_case → "q" letter_in_lower_case → "r" letter_in_lower_case → "s" letter_in_lower_case → "t" letter_in_lower_case → "u" letter_in_lower_case → "v" letter_in_lower_case → "w" letter_in_lower_case → "x" letter_in_lower_case → "y" letter_in_lower_case → "z"	letter_in_lower_case(1: "a") → "a" letter_in_lower_case(1: "b") → "b" letter_in_lower_case(1: "c") → "c" letter_in_lower_case(1: "d") → "d" letter_in_lower_case(1: "e") → "e" letter_in_lower_case(1: "f") → "f" letter_in_lower_case(1: "g") → "g" letter_in_lower_case(1: "h") → "h" letter_in_lower_case(1: "i") → "i" letter_in_lower_case(1: "j") → "j" letter_in_lower_case(1: "k") → "k" letter_in_lower_case(1: "l") → "l" letter_in_lower_case(1: "m") → "m" letter_in_lower_case(1: "n") → "n" letter_in_lower_case(1: "o") → "o" letter_in_lower_case(1: "p") → "p" letter_in_lower_case(1: "q") → "q" letter_in_lower_case(1: "r") → "r" letter_in_lower_case(1: "s") → "s" letter_in_lower_case(1: "t") → "t" letter_in_lower_case(1: "u") → "u" letter_in_lower_case(1: "v") → "v" letter_in_lower_case(1: "w") → "w" letter_in_lower_case(1: "x") → "x" letter_in_lower_case(1: "y") → "y" letter_in_lower_case(1: "z") → "z"	A = "A"; B = "B"; C = "C"; D = "D"; E = "E"; F = "F"; G = "G"; H = "H"; I = "I"; J = "J"; K = "K"; L = "L"; M = "M"; N = "N"; O = "O"; P = "P"; Q = "Q"; R = "R"; S = "S"; T = "T"; U = "U"; V = "V"; W = "W"; X = "X"; Y = "Y"; Z = "Z"; letter_in_lower_case = a b c d e f g h i j k l m n o p q r s t u v w x y z;	A = "A"; B = "B"; C = "C"; D = "D"; E = "E"; F = "F"; G = "G"; H = "H"; I = "I"; J = "J"; K = "K"; L = "L"; M = "M"; N = "N"; O = "O"; P = "P"; Q = "Q"; R = "R"; S = "S"; T = "T"; U = "U"; V = "V"; W = "W"; X = "X"; Y = "Y"; Z = "Z"; letter_in_lower_case = a b c d e f g h i j k l m n o p q r s t u v w x y z;	\\
letter_in_upper_case = "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K" "L" "M" "N" "O" "P" "Q" "R" "S" "T" "U" "V" "W" "X" "Y" "Z";	letter_in_upper_case = "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K" "L" "M" "N" "O" "P" "Q" "R" "S" "T" "U" "V" "W" "X" "Y" "Z";	letter_in_upper_case → "A" letter_in_upper_case → "B" letter_in_upper_case → "C" letter_in_upper_case → "D" letter_in_upper_case → "E" letter_in_upper_case → "F" letter_in_upper_case → "G" letter_in_upper_case → "H" letter_in_upper_case → "I" letter_in_upper_case → "J" letter_in_upper_case → "K" letter_in_upper_case → "L" letter_in_upper_case → "M" letter_in_upper_case → "N" letter_in_upper_case → "O" letter_in_upper_case → "P" letter_in_upper_case → "Q" letter_in_upper_case → "R" letter_in_upper_case → "S" letter_in_upper_case → "T" letter_in_upper_case → "U" letter_in_upper_case → "V" letter_in_upper_case → "W" letter_in_upper_case → "X" letter_in_upper_case → "Y" letter_in_upper_case → "Z"	letter_in_upper_case(1: "A") → "A" letter_in_upper_case(1: "B") → "B" letter_in_upper_case(1: "C") → "C" letter_in_upper_case(1: "D") → "D" letter_in_upper_case(1: "E") → "E" letter_in_upper_case(1: "F") → "F" letter_in_upper_case(1: "G") → "G" letter_in_upper_case(1: "H") → "H" letter_in_upper_case(1: "I") → "I" letter_in_upper_case(1: "J") → "J" letter_in_upper_case(1: "K") → "K" letter_in_upper_case(1: "L") → "L" letter_in_upper_case(1: "M") → "M" letter_in_upper_case(1: "N") → "N" letter_in_upper_case(1: "O") → "O" letter_in_upper_case(1: "P") → "P" letter_in_upper_case(1: "Q") → "Q" letter_in_upper_case(1: "R") → "R" letter_in_upper_case(1: "S") → "S" letter_in_upper_case(1: "T") → "T" letter_in_upper_case(1: "U") → "U" letter_in_upper_case(1: "V") → "V" letter_in_upper_case(1: "W") → "W" letter_in_upper_case(1: "X") → "X" letter_in_upper_case(1: "Y") → "Y" letter_in_upper_case(1: "Z") → "Z"	a = "a"; b = "b"; c = "c"; d = "d"; e = "e"; f = "f"; g = "g"; h = "h"; i = "i"; j = "j"; k = "k"; l = "l"; m = "m"; n = "n"; o = "o"; p = "p"; q = "q"; r = "r"; s = "s"; t = "t"; u = "u"; v = "v"; w = "w"; x = "x"; y = "y"; z = "z"; letter_in_upper_case = A B C D E F G H I J K L M N O P Q R S T U V W X Y Z;	a = "a"; b = "b"; c = "c"; d = "d"; e = "e"; f = "f"; g = "g"; h = "h"; i = "i"; j = "j"; k = "k"; l = "l"; m = "m"; n = "n"; o = "o"; p = "p"; q = "q"; r = "r"; s = "s"; t = "t"; u = "u"; v = "v"; w = "w"; x = "x"; y = "y"; z = "z"; letter_in_upper_case = A B C D E F G H I J K L M N O P Q R S T U V W X Y Z;	\\
				STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) (!({qi::alpha qi::char ("_"))});	STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) (!({qi::alpha qi::char ("_"))});	\\
				BOUNDARIES = (BOUNDARY >> *(BOUNDARY) NO_BOUNDARY);	BOUNDARIES = (BOUNDARY >> *(BOUNDARY) NO_BOUNDARY);	\\
				BOUNDARY = BOUNDARY_SPACE BOUNDARY_TAB BOUNDARY_CARRIAGE_RETURN BOUNDARY_LINE_FEED BOUNDARY_NULL;	BOUNDARY = BOUNDARY_SPACE BOUNDARY_TAB BOUNDARY_CARRIAGE_RETURN BOUNDARY_LINE_FEED BOUNDARY_NULL;	\\
				BOUNDARY_SPACE = " ";	BOUNDARY_SPACE = " ";	\\
				BOUNDARY_TAB = "\t";	BOUNDARY_TAB = "\t";	\\
				BOUNDARY_CARRIAGE_RETURN = "\r";	BOUNDARY_CARRIAGE_RETURN = "\r";	\\
				BOUNDARY_LINE_FEED = "\n";	BOUNDARY_LINE_FEED = "\n";	\\
				BOUNDARY_NULL = "\0";	BOUNDARY_NULL = "\0";	\\
				NO_BOUNDARY = "";	NO_BOUNDARY = "";	\\
						\\
						\\

						<pre>\ { LA_IS, { T_NAME_0 }, { "program____part1", {\ { LA_IS, { "" }, 7, { T_NAME_0, "program_name", T_SEMICOLON_0, T_BODY_0, T_DATA_0, "declaration_optional", T_SEMICOLON_0 }}\ }}\ \ }\ "program_rule"</pre>
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